

See discussions, stats, and author profiles for this publication at: <https://www.researchgate.net/publication/225238895>

Small Defects and Inhomogeneities in Fatigue Strength: Experiments, Models and Statistical Implications

Article in *Extremes* · January 1999

DOI: 10.1023/A:1009976418553

CITATIONS

206

READS

4,094

2 authors, including:



Stefano Beretta

Politecnico di Milano

292 PUBLICATIONS 6,811 CITATIONS

SEE PROFILE



Small Defects and Inhomogeneities in Fatigue Strength: Experiments, Models and Statistical Implications

Y. MURAKAMI

Department of Mechanical Science and Engineering, Kyushu University, Fukuoka, Higashi-ku, 812-81 Japan

S. BERETTA

Politecnico di Milano, Dipartimento di Meccanica, Piazza Leonardo da Vinci 32, I-20133 Milano, Italy

[Received May 4, 1999; Revised September 7, 1999; Accepted September 8, 1999]

Abstract. The method explained in this paper for quantitative evaluation of fatigue limit for materials containing defects is based on the experimental evidences that inhomogeneities and micro-notches can be treated like cracks.

First, the basic concept of the $\sqrt{\text{area}}$ parameter model is explained introducing the various data obtained by the first author's group for over last 15 years. Evidences are shown that small cracks, defects and nonmetallic inclusions having the same value of the square root of projection area, $\sqrt{\text{area}}$, have the identical influence on the fatigue limit regardless of different stress concentration factors. Various applications of these concepts to various defect types and microstructural inhomogeneities are shown.

Since the estimation of fatigue strength is related to the estimation of the size of maximum defects occurring in a piece, the methods for searching the defects and the quality control of materials with respect to inclusion or defect rating as well as their statistical implications are discussed.

Key words. Fatigue limit, maximum defect, projection area, inclusion, extremes, Gumbel

AMS 1991 Subject Classifications. 60G70, 62N05, 60D05

List of symbols used

$\sqrt{\text{area}}$	square root of the defect area projected onto a plane perpendicular to the applied stress
$\sqrt{\text{area}}_{\text{crit}}$	critical dimension below which a defect does not affect fatigue strength
ΔK	stress intensity factor range
ΔK_{th}	threshold stress intensity factor range
$\Delta K_{th,lc}$	threshold stress intensity factor range for a long crack
λ, δ	location and scale parameters of the Gumbel distribution
σ	applied stress
σ_w	fatigue limit in presence of a defect or a short crack
σ_{wo}	fatigue limit of the material (smooth specimens)
σ_{wl}	lower bound of fatigue limit in presence of inclusions
σ_{wu}	upper bound of the fatigue limit

h_{med}	average size of inclusions
HV	Vickers hardness of the metal matrix
POT	Peak over Threshold
R	stress ratio ($R = \sigma_{min}/\sigma_{max}$)
SIF	stress intensity factor
S_0	control area in extreme value sampling
V_c	volume of the component
V_0	control volume in extreme value sampling
$x(T)$	dimension of the defect with return period T

Background of the $\sqrt{\text{area}}$ parameter model

What is fatigue limit?

A specimen containing a small microhole that was subjected to fatigue stresses at a stress level corresponding to fatigue limit is shown in Figure 1(a) it can be observed the presence of cracks emanating from the artificial hole. Figure 1(b) shows the sketches of these cracks observed at the fatigue limit [1]. These cracks are so-called nonpropagating cracks which started from the hole edge in the early stages of stress cycle and afterward stopped propagating under the same stress level. Many similar observations have been given in the literature [1–4].

These experimental evidences support the concept that the fatigue limit of most materials is not the critical stress for crack initiation but the threshold stress for nonpropagation of the crack which emanated from original cracks, defects or inhomogeneities.

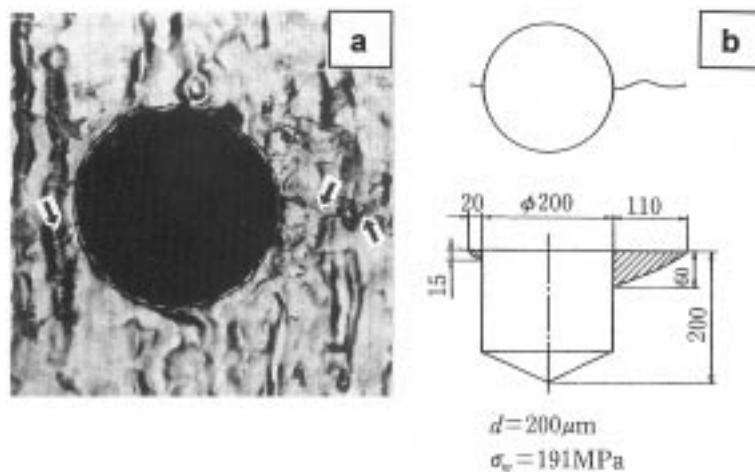


Figure 1. Nonpropagating cracks observed at fatigue limit of a 0.45% carbon steel: (a) surface of specimen (the arrows indicate the short crack); (b) shape of the nonpropagating cracks [1].

When 12 holes with $40\text{ }\mu\text{m}$ diameter were introduced on the surface of one specimen tested at the fatigue limit stress [1], some holes had two nonpropagating cracks at the points of stress concentration of the hole (see Figure 1) but other holes had only one or no nonpropagating crack. At stress 5 MPa higher than the fatigue limit all specimens failed, and no specimens had cracks at stress 5 MPa lower than the fatigue limit.

Thus, if the microstructure of materials is well controlled, fatigue limit is a phenomenon observed only in narrow band of applied stress. From this viewpoint, the scattered fatigue strength of high strength steels should not be attributed to their hard microstructures, but it is mostly caused by the presence of nonmetallic inclusions and contained in the hard microstructures.

Figure 2 shows the typical relationship between fatigue limit and dimension of the defects (the data depict a part of the so-called Kitagawa-Takahashi diagram [5]). The important experimental observation is that the fatigue limit of a piece containing a micro-notch increases as the dimension of the defects is decreased, and it becomes equal to the fatigue limit of smooth specimens σ_{w0} when the defects are smaller than a critical crack size $\sqrt{\text{area}_{\text{crit}}}$.

Since small cracks always develop at the tip of defects at stress levels near the fatigue limit, the defects can be treated as cracks. Thus, the fatigue limit can be predicted as the cyclic stress level at which these small cracks do not propagate. In the following we will expose the $\sqrt{\text{area}}$ model which, among many other models [3–4], is simple and easy to apply. Nevertheless the cases where this method is not valid will also be mentioned.

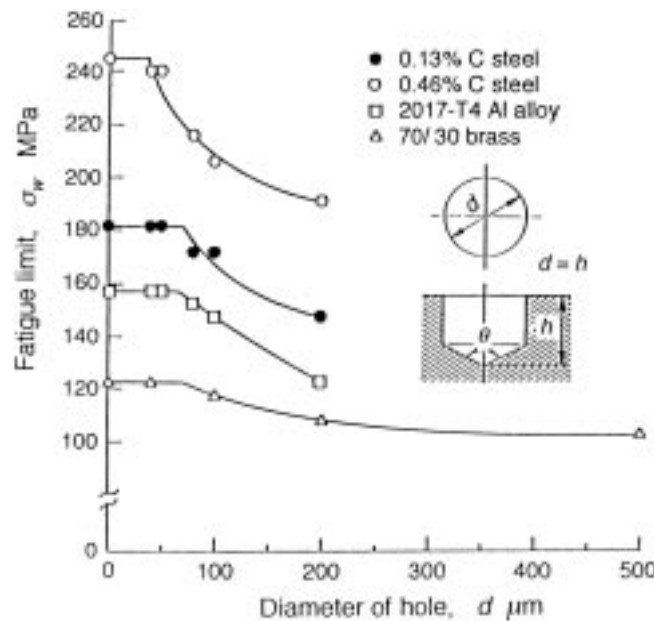


Figure 2. Experimental results on specimens containing micro-holes [3].

The $\sqrt{\text{area}}$ parameter model

The name of the model derives from the geometrical parameter $\sqrt{\text{area}}$ (square root of the defect area projected onto a plane perpendicular to applied stress) that allows calculating the maximum stress intensity factor (SIF) along the front of cracks with irregular shape [6]:

$$\Delta K = 0.65 \cdot \Delta \sigma \cdot \sqrt{\pi \cdot \sqrt{\text{area}}}. \quad (1)$$

A crack does not propagate when its stress intensity factor range ΔK is lower than a threshold value $-\Delta K_{th}$. Figure 3 shows the values of ΔK_{th} of various materials (stress ratio $R = -1$) as a function of crack size expressed in terms of the parameter $\sqrt{\text{area}}$: we can notice that ΔK_{th} is not constant for small cracks but it has a good correlation with $\sqrt{\text{area}}$ of defects. In particular:

$$\Delta K_{th} \propto \sqrt{\text{area}}^{1/3}. \quad (2)$$

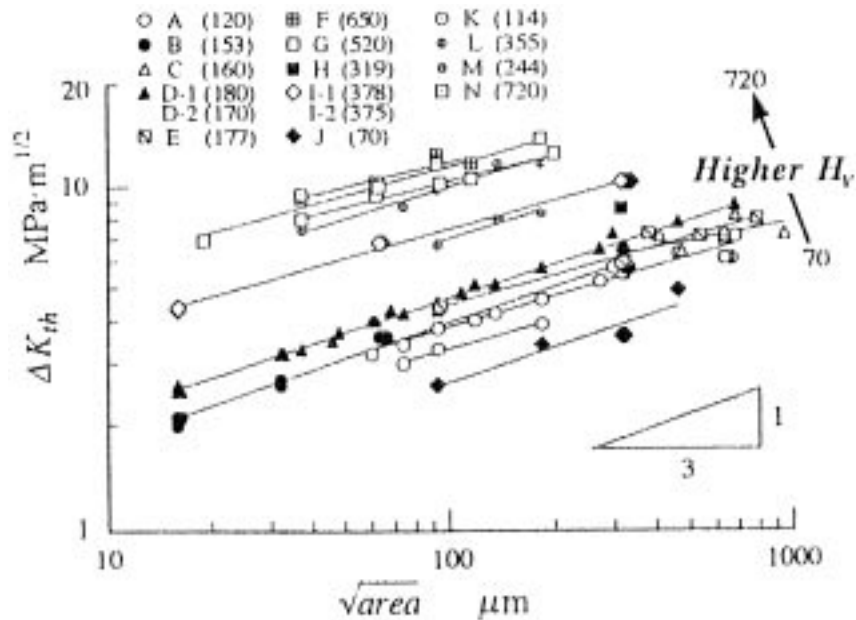


Figure 3. Relationship between ΔK_{th} and $\sqrt{\text{area}}$ for various defects and cracks (letters correspond to materials in [2] and numbers indicate the Vickers hardness) [2].

Looking at Figure 3 again, we can note that the data lines of ΔK_{th} for 15 materials are exactly ordered from lower to higher values of Vickers hardness (HV). Thus, applying the least square method to all the data, the resulting relationship was [2]:

$$\Delta K_{th} = 3.3 \cdot 10^{-3} \cdot (HV + 120) \cdot (\sqrt{\text{area}})^{1/3}. \quad (3)$$

By knowing from equation (3) the relationship between SIF and $\sqrt{\text{area}}$, equation (3) can be rewritten in terms of fatigue limit stress under alternating stresses ($R = -1$):

$$\sigma_w = 1.43 \cdot \frac{(HV + 120)}{\sqrt{\text{area}}^{1/6}} \quad (4)$$

where, the units are ΔK_{th} : MPa $\sqrt{\text{m}}$, HV: kgf/mm², $\sqrt{\text{area}}$: μm , σ_w : MPa. Considering that the geometrical parameter $\sqrt{\text{area}}$ was derived theoretically, we can say equation (4) is a semi-theoretical and semi-empirical formula.

In recent studies, [3,4] extended equation (4) to more general cases including R ratio as follows.

$$\text{surface defects:} \quad \sigma_w = 1.43 \cdot (HV + 120) / (\sqrt{\text{area}})^{1/6} \cdot [(1 - R)/2]^\alpha \quad (5)$$

$$\text{internal defects:} \quad \sigma_w = 1.56 \cdot (HV + 120) / (\sqrt{\text{area}})^{1/6} \cdot [(1 - R)/2]^\alpha \quad (6)$$

where:

$$\alpha = 0.226 + HV \cdot 10^{-4} \quad (7)$$

The $\sqrt{\text{area}}$ parameter model (expressed by equation (3–7)) is essentially valid in the cases where equation (3) is valid. For mild steels the equation (3) has resulted to be valid in the range $\sqrt{\text{area}}_{crit} - 1000 \mu\text{m}$ (see Figure 2(b)), from this dimension ΔK_{th} data tend to a constant value ΔK_{thlc} .

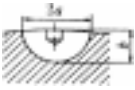
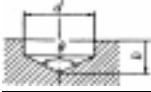
Few cases in which the $\sqrt{\text{area}}$ model is not valid are the components that contain relatively large defects in this transition region: in such cases the equation (3) is not valid and the ΔK_{th} behavior can be kept constant or described with other models [3].

Applications of the $\sqrt{\text{area}}$ parameter model to artificial defects and nonmetallic inclusions

Holes and cracks

From the viewpoint of examining the $\sqrt{\text{area}}$ parameter model, the comparison of fatigue limit of specimens containing an artificial drilled hole and an artificial surface crack

Table 1. Comparison of rotating bending fatigue limit containing small artificial holes and defects (annealed 0.46% carbon steel) [4].

Shape of initial hole and crack	$\sqrt{\text{area}} \mu\text{m}$	Experimental value of fatigue limit, σ_w MPa
 $2a = 213 \mu\text{m}$ Aspect ratio: $b/a = 0.7$	112	200.9
 $d = 200 \mu\text{m}$ $h = 100 \mu\text{m}$ $\theta = 120$	115	200.9

Note. Experiments were carried out with steps of 0.5 kg/mm^2 : the fatigue limits resulted to be 20.5 kg/mm^2 .

having identical values of $\sqrt{\text{area}}$ is interesting. Table 1 shows the shape of the initial hole and crack [4]. The crack was introduced by fatigue test using a specimen containing a smaller artificial hole with the diameter of $40 \mu\text{m}$. The specimen was annealed after preliminary fatigue test to relieve the prior fatigue history. As shown in Table 1, the fatigue limits measured with these two kinds of specimens are exactly the same regardless of the big difference in stress concentration factor of hole and crack.

To examine more in detail the validity of the $\sqrt{\text{area}}$ parameter model, specimens of a hard steel (maraging steel: $\text{HV} = 510$) were tested by tension-compression fatigue [8]. As shown in Figure 4, the fatigue limits of the specimen having an initial crack (C) and a two-hole defect (B) with fatigue crack introduced by preliminary fatigue test are the same (402 MPa) for the same value of $\sqrt{\text{area}}$, though that for the specimen containing one hole (A) is 421 MPa which is only 4.7% higher than (B) and (C).

Nonmetallic inclusions at fracture origin

It is quite common that a white circular mark named a fish-eye is observed on the fatigue fracture surface of high-strength steels. Usually a nonmetallic inclusion exists at the center of a fish-eye. Figure 5 shows the typical nonmetallic inclusion at fish-eye [9]. The fatigue fracture origin of high-strength steels is mostly at nonmetallic inclusions such as MnS , Al_2O_3 , SiO_2 , and TiN having various shapes and sizes. In terms of fatigue limit, we can consider nonmetallic inclusions to be mechanically equivalent to stress-free defects [4].

When nonmetallic inclusions cause fatigue failure, it is very difficult to predict the fatigue strength in advance of fatigue test. We can identify the size and location of the fatal inclusion on the fracture surface of specimen after fatigue fracture. In such cases, the stress applied at the fatal inclusion should be larger than the value that is calculated by the fatigue strength prediction equations (5) or (6). Table 2 shows the comparison between such experimental results and the calculated values for the case of a bearing steel [10]. We can see in Figure 6 that the stresses (σ') applied at the fatal inclusions are actually larger than the estimated fatigue limits (σ'_w).

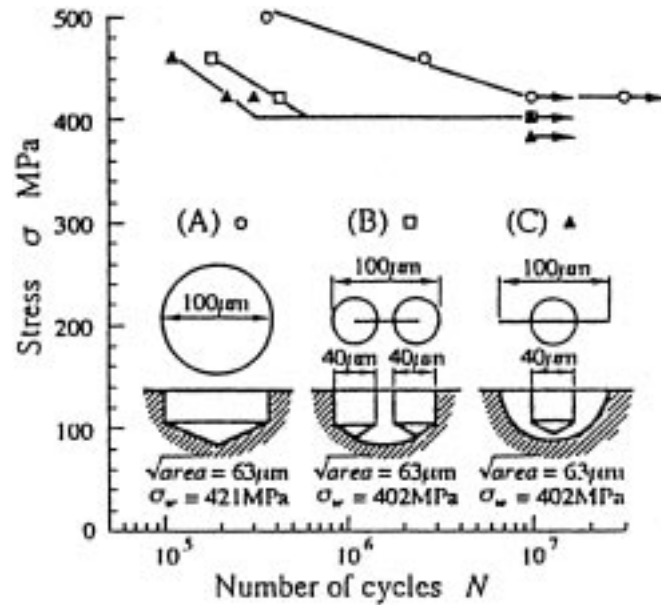


Figure 4. S-N curves for a maraging steel with holes and crack having the same $\sqrt{\text{area}}$ [8].

Shape of inclusions

The experimental facts on holes and cracks shown in Table 1 and Figure 6 suggest that there should be no difference in the influence of inclusion shape if the values of $\sqrt{\text{area}}$ of two inclusions are the same. This prediction is completely agreement with the results of the fatigue tests conducted by [11] who artificially introduced spherical and angular

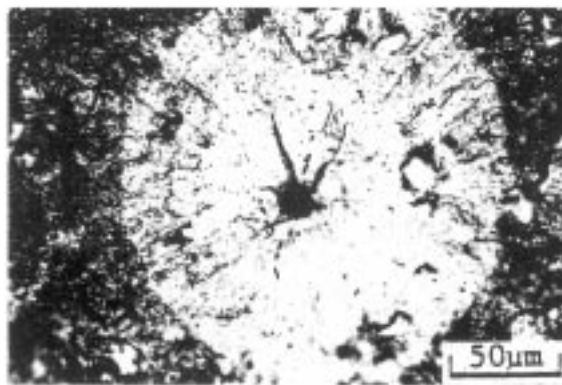


Figure 5. Typical example of fisheye: the origin of fatigue fracture is the inclusion at the center [9].

Table 2. Size and location of inclusions and predicted fatigue limits (rotating bending fatigue tests, bearing steel) [10].

Nominal stress at surface, σ (MPa)	Cycles to failure, N_f	Inclusion size, $\sqrt{\text{area}}(\mu\text{m})$	Distance from surface, $h(\mu\text{m})$	Shape of inclusions	Nominal stress at inclusion, σ' (MPa)	Fatigue limit predicted by Eq.(5), σ'_w (MPa)	σ'/σ'_w
981	50.11×10^4	37.5	80		961	686	1.40
981	39.21	29.3	170		941	715	1.32
932	683.38	32.5	570		807	703	1.15
883	138.24	53.5	200		841	647	1.30
981	75.06	31.0	350		900	708	1.27
932	160.35	25.6	240		878	732	1.20
932	11.12	47.0	100		909	661	1.38
932	23.40	100.7	1030		706	582	1.21
834	420.00	69.9	600		715	619	1.16

alumina particles into ingots and found no definite difference of the effect of inclusion shape. Their experimental data can be also arranged clearly by the $\sqrt{\text{area}}$ parameter model [4].

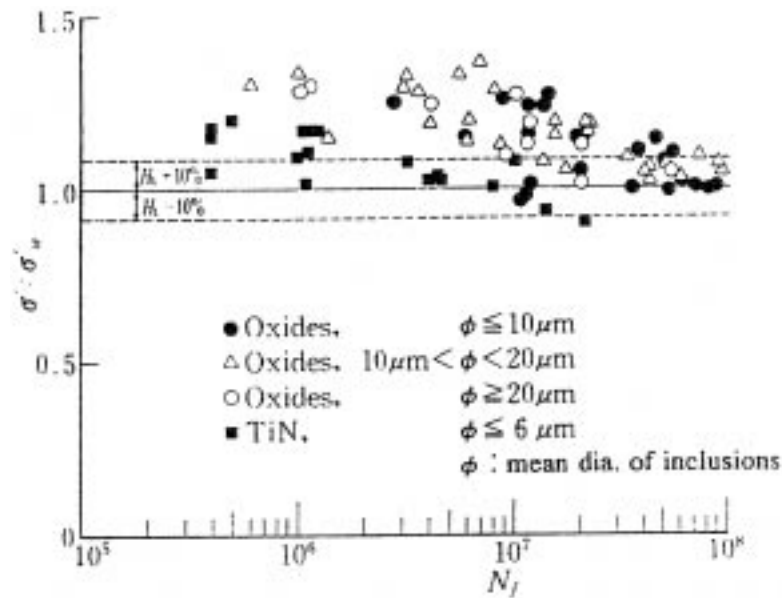


Figure 6. Comparison between the failure stress σ' and the predicted fatigue strength σ'_w (rotating bending fatigue, bearing steel) [10].

Prediction of scatter of fatigue strength due to nonmetallic inclusions

Murakami et al. [10] and Murakami and Usuki [12] proposed a method of predicting the upper and the lower bound of fatigue strength. The upper bound of fatigue strength corresponds to σ_{wu} when defects or inclusions do not affect fatigue fracture, and its value can be empirically estimated as [4]:

$$\sigma_{wu} = 1.6 \text{ HV} \pm 0.1 \text{ HV} \quad (\text{HV} < 400). \quad (8)$$

The lower bound σ_{wl} of fatigue strength can be obtained when a large inclusion is located in contact with the surface of a specimen [12]. The prediction equation is:

$$\sigma_{wl} = 1.41 \cdot (\text{HV} + 120) / (\sqrt{\text{area}_{\max}})^{1/6} \quad (9)$$

where $\sqrt{\text{area}_{\max}}$ is the expected maximum size of inclusions contained in a definite volume.

In order to estimate $\sqrt{\text{area}_{\max}}$ Murakami and his colleagues [4,13,14] applied the statistics of extreme values [15] to the distribution of inclusions. Figure 7 shows the distribution of the maximum size $\sqrt{\text{area}_{\max}}$ of inclusions observed in the unit area $S_0 (= 0.0309 \text{ mm}^2)$ of a spring steel (Japanese Industrial Standard SUP12). Estimating the return period T for N specimens, the characteristic largest size of inclusions $\sqrt{\text{area}_{\max}}$ can be predicted by the procedure indicated in the figure (see section Statistical Analysis of Defects).

Introducing HV and $\sqrt{\text{area}_{\max}}$ into equation (9), the lower bound σ_{wl} of fatigue strength for N specimens is predicted as in Figure 8. The lower bound of experimental data is well predicted by the equation. Similar applications to various materials have been reported in [4,16–22].

Applications to other material defects*Graphite spheroids and defects in cast irons*

Nodular cast iron contains numerous small defects as graphite nodules in the structure (see Figure 9). Since the fatigue strength is strongly affected by the nodules or casting defects, a number of researches [23–30] has been recently dealt with the successful application of $\sqrt{\text{area}}$ parameter model to the fatigue strength evaluation of this material.

Figure 10 shows the distribution of nodule size ($\sqrt{\text{area}}$) of the material used in the fatigue test by [23]. Table 3 demonstrates the excellent agreement between the experimental results and the prediction by the $\sqrt{\text{area}}$ parameter model.

When the size ($\sqrt{\text{area}}$) of other defects is larger than that of graphite nodules, the fatigue limit of is determined by the defects with the maximum size. Beretta et al. [31] analyzed conrods made of nodular cast iron and they found defects constituted by

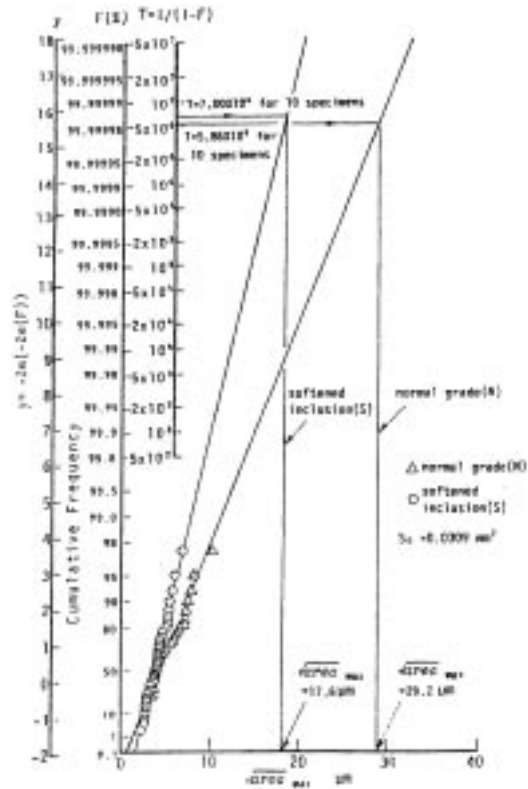


Figure 7. Cumulative frequency of the extreme values of inclusions in spring steel (S_0 : standard control area) [9].

inclusions and pores (Figure 10). Since the estimated maximum defects exceeded 1 mm, the fatigue limit was predicted assuming that ΔK_{th} was constant.

Pores in cast aluminum alloy

The application of aluminum alloys to automobile industry is very active. However, it is well known that pores contained in cast aluminum alloys causes the low fatigue strength and the scatter of fatigue strength.

Kobayashi and Matsui [32] confirmed first the availability of the $\sqrt{\text{area}}$ parameter model to an aluminum alloy (JIS-AC4C, ASTM356.0) and then applied the same method to the strength design of a rocker arm for truck engines. Figure 11 shows the size distribution of the Al alloys and the S-N curve of the rocker arm with the predicted lower bound of the fatigue limit for 100 parts.

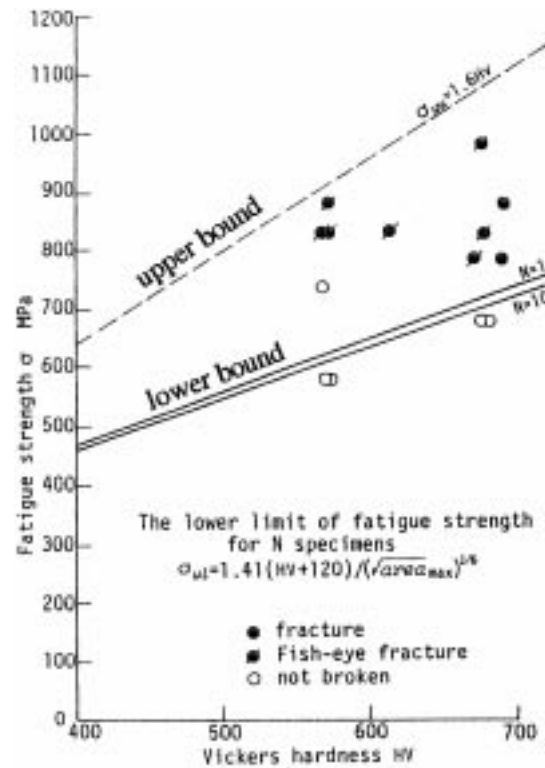


Figure 8. Comparison of the predicted lower limit of fatigue strength with the experimental results (softened inclusions in spring steel) [9].

Second phases in Al-Si eutectic alloy

The inhomogeneity phases such as Si and Al-Fe-Mn in Al-Si eutectic alloys behave like hard nonmetallic inclusions in fatigue, because the hardness of these phases is HV 700 while that of aluminum matrix ranges from 100 to 170. Figure 12 shows the microstructure of Al-Si eutectic alloy. Fatigue cracks initiate at the interface between these hard second phases and the soft matrix. Thus, the same concept of fatigue strength prediction for nonmetallic inclusions has been successfully applied to this material [33].

Carbides in tool steel

Tool steels contain various carbides and they are usually formed at high temperature to avoid carbide cracking. Large carbides cause fatigue failure like nonmetallic inclusions and it is expected that the $\sqrt{\text{area}}$ parameter model will be successfully even if carbides are cracked or not cracked before fatigue test.

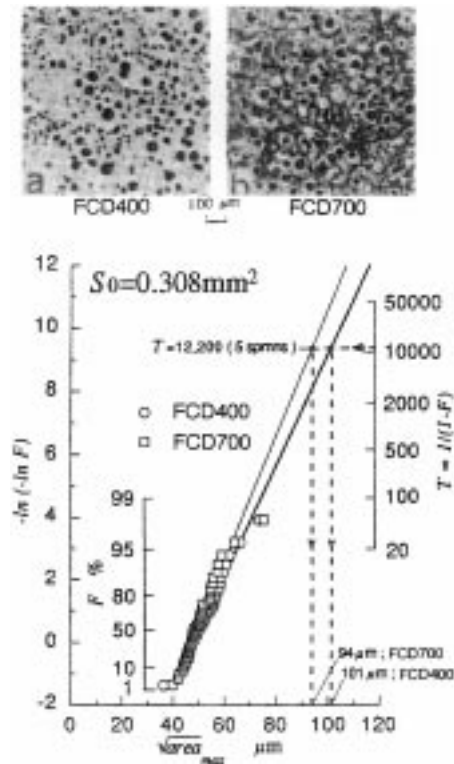


Figure 9. Distribution of maximum sizes of graphite nodules for two nodular cast irons [27].

Natsume, Muramatsu and Miyamoto [34] conducted interesting experiments from this expectation. They prepared specimens containing uncracked carbides by high temperature forming and others containing cracked carbides by room temperature forming (the hardness of the matrix was adjusted to be identical in both specimens). They found no difference in fatigue limit of these two materials, see Figure 13, and both the fatigue limits were in good agreement with the value predicted by equation (9). This result may be against conventional knowledge but it was naturally predicted by the $\sqrt{\text{area}}$ parameter model.

Quality control of materials by defect rating based on statistics of extremes

Extreme defects

From Figure 2 we can see that for a given stress level the failure or survival of a piece is determined by the size of the largest crack or defect which is present. As a consequence of this fact the fatigue strength is controlled by the dimension of maximum occurring defects

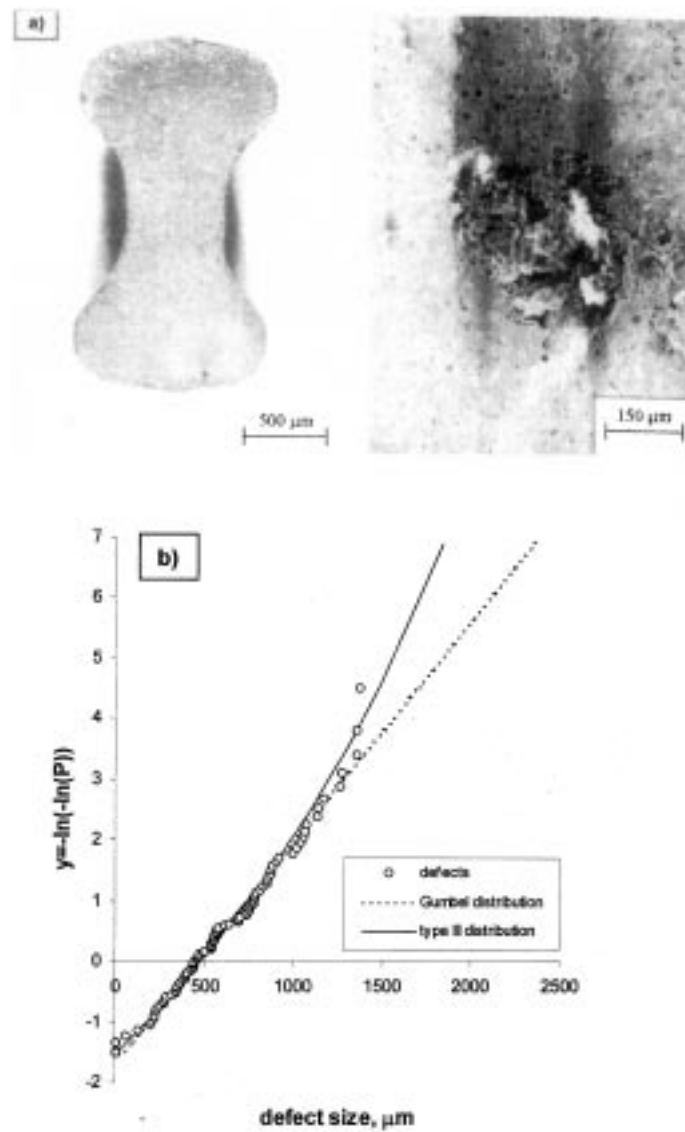


Figure 10. Conrods made of nodular cast iron: (a) a defect at the origin of fatigue failure; (b) distribution of maximum defects [31].

rather than the average dimension of inhomogeneities: the most suitable statistical method is the analysis of extremes based on extreme value sampling.

The detection of defects with an extreme value sampling can be carried out by collecting the defects at the origin of fatigue failures [3–4] or by adopting the method of polished sections [4,9,11–14].

Table 3. Comparison of rotating bending fatigue tests on nodular cast irons and predictions with $\sqrt{\text{area}}$ parameter model [23].

Material	$\sqrt{\text{area}}_{\text{max}} (\mu\text{m})$	$\sigma_{w0,\text{pred}} (\text{MPa})$	$\sigma_{w0,\text{exp}} (\text{MPa})$
FCD400	101	202	220
FCD700	94	274	280

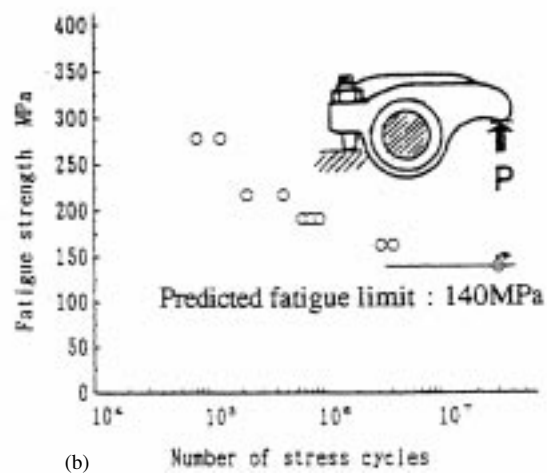
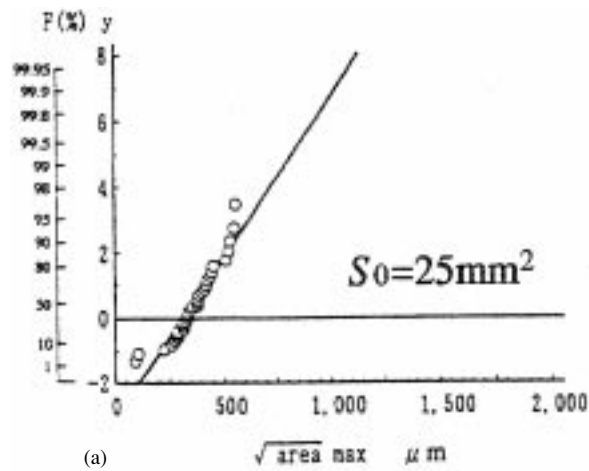


Figure 11. Rocker arm made of cast Al alloy: (a) distribution of extreme values for pore size; (b) S-N curve [32].

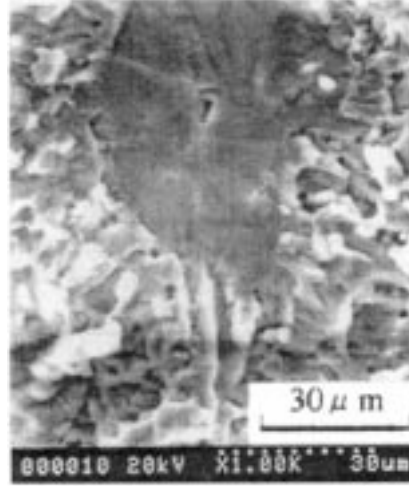


Figure 12. Si-phase at the origin of fatigue failure of an extruded Al alloy (high cycle fatigue test, $\sigma = 196$ MPa, $N_f = 7.10^6$) [33].

The first method is the most reliable (a specimen breaks at the largest defect) but it also implies a number of expensive and time consuming operations (specimens machining and preparation, fatigue tests, microscope observations) that in general can be done only for research tests or as the final validation for delivering a material or a component.

The second method is based on cutting the piece or the material in planes perpendicular to the direction of applied stress and observing under microscope the cut surfaces after polishing. The sampling of inhomogeneities is done by recording the maximum defect size in a given control area S_0 .

Statistical analysis of defects

It has been widely demonstrated by Murakami and co-workers [4,11–14] that the defect data obtained with such procedures can be successfully described using the Largest Extreme Value Distribution (LEVD) [15] in order to calculate the dimension of the maximum occurring defect.

Let $F(x)$ be the LEVD distribution [15]:

$$F(x) = \exp \left\{ - \exp \left[- \frac{(x - \lambda)}{\delta} \right] \right\} \quad (10)$$

where λ and δ are referred to as location and scale parameter respectively.

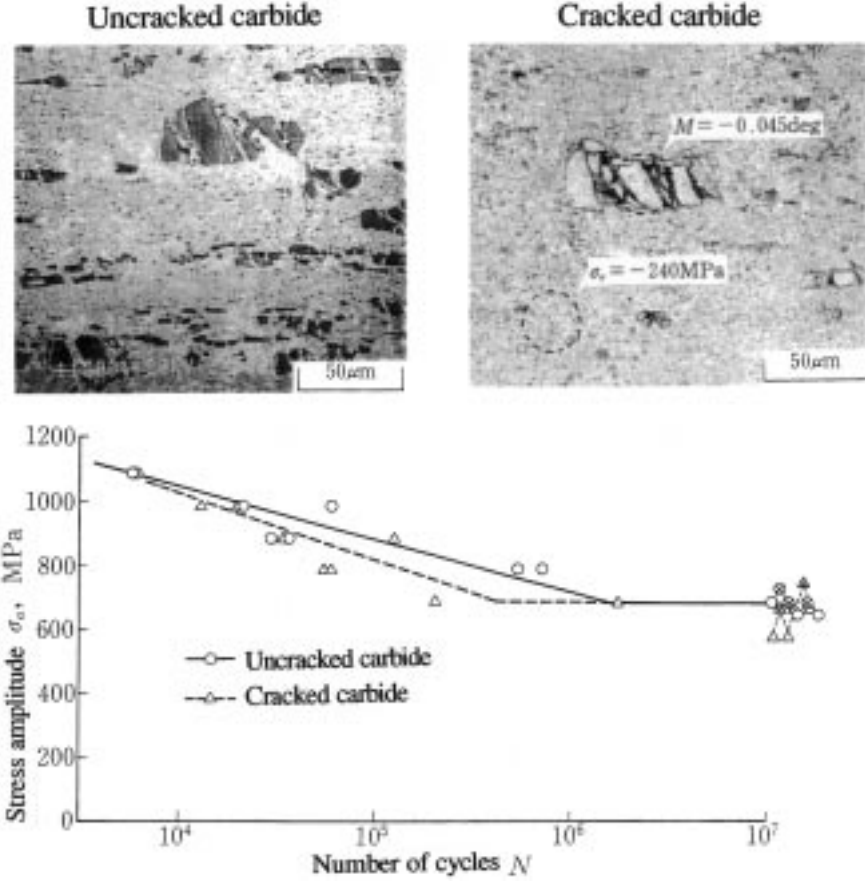


Figure 13. S-N curves of tool steel containing cracked or uncracked carbides [34].

The P^{th} quantile of the distribution is:

$$F^{-1}(P) = \lambda - \delta \cdot \ln(-\ln P). \quad (11)$$

If V_c is the volume of a component and V_o is the control volume used in defect sampling then the characteristic largest defect in the component (the defect which is expected to be exceeded once in the volume V_c) has a return period $T = V_c/V_o$. It can be calculated as:

$$x(T) = \lambda - \delta \cdot \ln \left[-\ln \left(1 - \frac{1}{T} \right) \right]. \quad (12)$$

A problem that has to be taken into account is the uncertainty of the estimates of $x(T)$.

The simplest approach is treating the confidence band of the defect with return period T

as the range of the possible outcomes of the true value $x(T)$. So fatigue strength calculations can be done by considering the dimension of the defect corresponding to the upper bound of the 95% confidence interval of $x(t)$ [37] (see Figure 14). However, this approach is correct only under the hypothesis that the bias of $x(T)$ estimates is negligible.

Beretta and Murakami [35] proposed an alternative and more precise method, which analyses the statistical properties (bias, efficiency), of the three different LEVD estimators most widely used: Least Squares [13,14], Moment and Maximum Likelihood methods [35].

The analysis, carried out with Montecarlo simulation, showed that from sample sizes of 30 defects the maximum likelihood (ML method) is the most efficient estimator and the confidence of its estimates correspond to the Cramer-Rao lower bound.

By considering the statistical properties of ML estimates Beretta and Murakami found the minimum number of defects to be examined so that the bias becomes stable and the width of error band is lower than the precision of the fatigue strength model adopted. These results are summarized in maps that allow the user to easily calculate the sample size needed to estimate the maximum defect with a precision suitable for evaluating the fatigue strength (Figure 15).

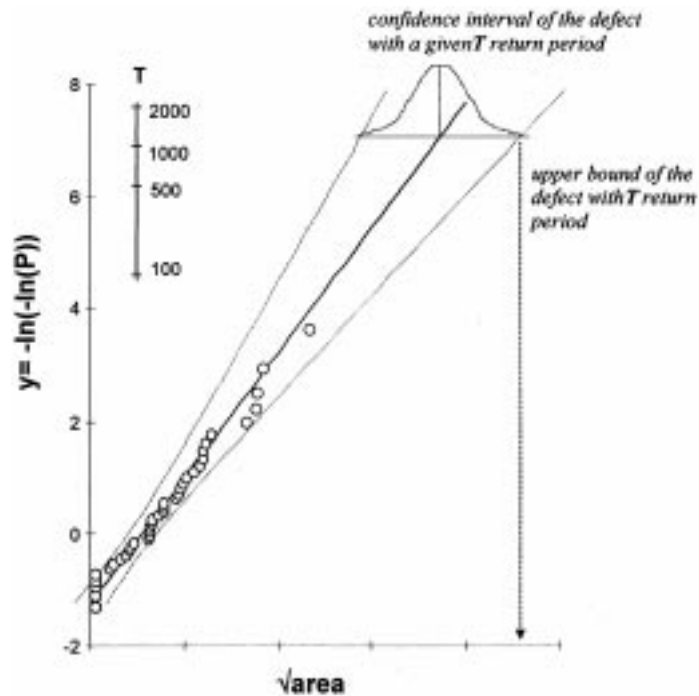


Figure 14. Simple treatment for the error in estimation of extreme defect.

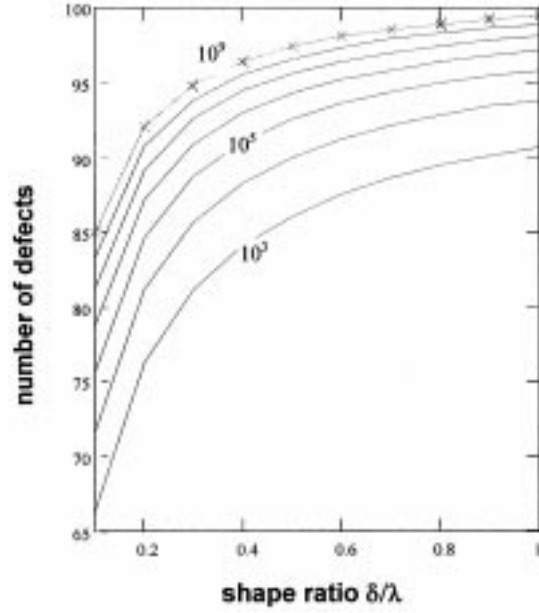


Figure 15. Maps for minimizing the bias of the maximum likelihood estimate of a defect with return period T [35].

Remarks and discussion

There are a number of open problems in the application of these methods and it is worth focusing them for the sake of a better co-operation and comprehension among material scientists, engineers and statisticians.

Control volume

The first problem that is met in using this approach is how to apply equation (12) for estimating maximum defects in a 3-D volume V_c if the defects were searched on a polished 2-D surface with a control area S_0 .

A solution to this problem was found by [36] who analyzed the problem with numerical simulation under the hypothesis of random cuts in a volume containing spheres with an exponential distribution (Figure 16). They suggested defining a conventional control volume V_0 as:

$$V_0 = S_0 \cdot h_{\text{med}} \quad (13)$$

where h_{med} is the average size of the extreme particles recorded in 2-D sampling.

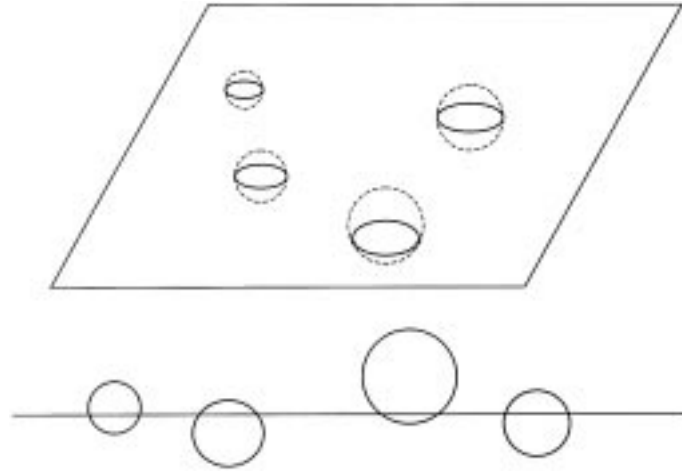


Figure 16. Sketch of the simulation of polished sections adopted by [36].

Takahashi and Shibuya [37], analyzing a theoretical model of Wicksell's problem under the hypothesis that the size distribution of random spheres follows a generalized gamma, showed that the approximation of equation (13) is near the theoretical results.

However, there is a series of typical problems which do not fall under the hypothesis of [38]: (i) no investigation has been carried out for other types of particle distributions; (ii) most of the defects of forged components and cold drawn wires are very thin subsurface microfolds similar to cracks (see Figure 17) [31,38,39]; (iii) in the case of cast nodular iron the defects are clustered spheroids whose average dimension is higher than the average dimension of spheroids; (iv) in general there are several types of defects with different density.

Type I, II or III extreme distribution?

A problem that often generates confusion about the application of the statistics of extremes to defects is the choice of the Gumbel or type I distribution, in particular the fact that the right tail is unbounded (the absence of lower limit is not a problem in estimating large defects).

Obviously there should exist an upper limit to the dimension of defects in a piece (at least the defects cannot be bigger than the transverse dimension of the piece itself) however there are no direct evidences that this inherent upper limit could influence the current analyses which are based on small areas (or volumes).

The few results of extensive measurements of inclusion sizes tend to show that the inclusions follow exponential [40] or lognormal [41,42] distributions: this supports the use of type I distribution for extreme defects. Moreover the huge amount of fatigue test results obtained by Murakami (with specimen volumes in the range 20–1000 mm³) in these

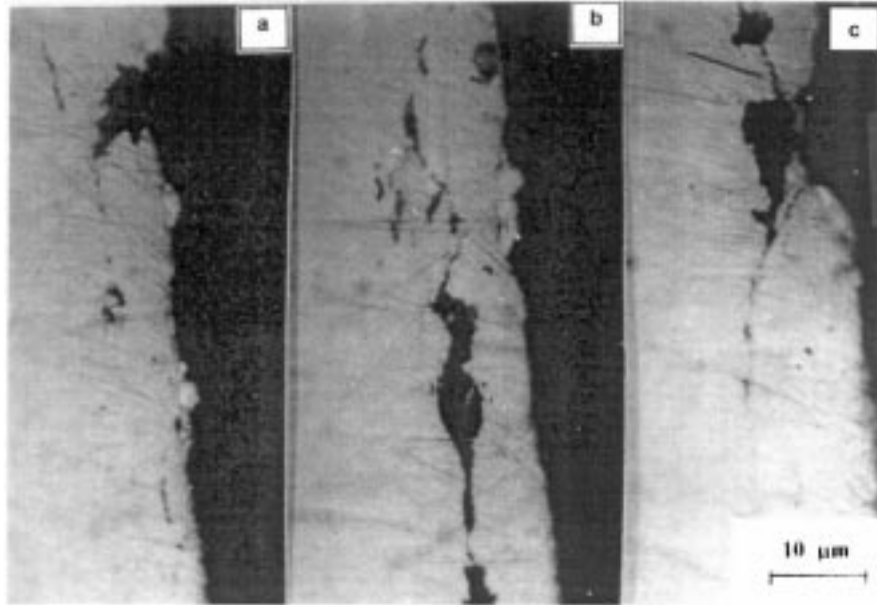


Figure 17. Defects found on longitudinal polished section of cold drawn wires: (a) cavity; (b) fold; (c) triple point [38].

recent years [4,8–10,12,19–22] show that inclusions at fracture origin follow a Gumbel distribution.

Calculating the fatigue strength of a big volume by estimating the dimension of the defects present in the volume is one of the ways that can be used for treating the “statistical fatigue size effect”.³ The references on statistical size effect show that this scaling effect is likely only in the case of materials containing real defects [43–47]. However, the volume of specimens used in these experiences is similar to the one of the Murakami’s experiments and so unfortunately no information about the existence of an asymptotic upper limit can be inferred from these data.

In the case of cast components the problem can be more easily solved. The upper limit of defect dimension is the thickness of the component and the defect analysis could be done using a type III distribution with this *a priori* upper limit. In the case of the cast iron conrods, the estimation of lower bound of fatigue strength for a batch of 1000 pieces is approximately 10% higher than the prediction based on Gumbel distribution (see Figure 11).

Sampling and image analysis

Many advanced statistical methods [38,48] need information, such as average number or distribution of the whole particles, that are apparently “simple” but that in reality are unlikely to be obtained. For example if we observe Figure 18, where a polished section is

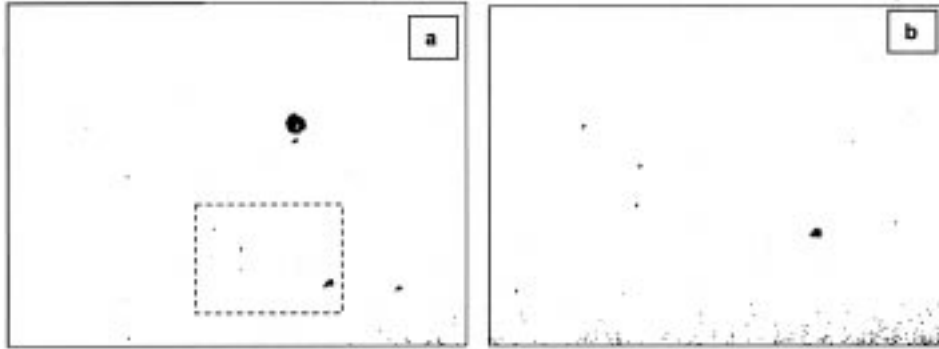


Figure 18. Defects on a polished section of a SCM35 steel under optical microscope: (a) observation at $100\times$; (b) observation at $200\times$ of the dashed control area in (a).

observed at different magnifications, we can notice two different facts: i). the higher the magnification, the higher the number of particles we can see (this process seems almost unbounded); ii). what we are seeing at high magnifications is a type of defects which is likely to be different from the big defect in Figure 18(a).

These problems, which suggest maximizing the control area S_0 , are related to the well-known presence of different defect types. From this point of view the possible solutions appear to be the use of POT methods [49] or to calculate the “density” of the most detrimental defects for choosing an appropriate control area [50,51].

This means that the result of defect analysis can be greatly influenced by the detection and analysis methods adopted and a standard procedure for “Extreme Value Inclusion Rating” should be established.

Conclusions

Experimental observations show that fatigue limit is the threshold condition for crack propagation, and therefore the stress intensity factor for the crack controls the critical condition rather than stress concentration factor of defects and other inhomogeneities. The conclusions are summarized as follows.

1. Fatigue problems of very small defects and nonmetallic inclusions cannot be solved by the conventional fatigue notch factor approach and the key point is to correctly consider the threshold condition of very small defects and nonmetallic inclusions.
2. The fatigue limit of specimens or machine components containing a small defect or crack is dependent on the Vickers hardness (HV) of the microstructure and the square root of the projection area of the defect or crack, $\sqrt{\text{area}}$.

3. The $\sqrt{\text{area}}$ parameter model can be also applied to inclusion problem in fatigue, because the effect of a nonmetallic inclusion is mechanically equivalent to that of a small defect. The fatigue fracture from a nonmetallic inclusion occurs when the applied stress at the inclusion exceeds the value calculated by the fatigue strength prediction equation for small defects.
4. The scatter of fatigue strength of hard steels is caused by the fracture from nonmetallic inclusions. Only the upper and lower bound of fatigue strength σ_{wu} and σ_{wl} can be estimated: σ_{wu} is empirically proportional to HV, σ_{wl} can be estimated using the statistics of extreme values which enables one to estimate the maximum size $\sqrt{\text{area}}_{\text{max}}$ of inclusion contained in many specimens and machine components.
5. There are some cases where the $\sqrt{\text{area}}$ parameter model cannot be directly applied because the defect sizes fall in the region where ΔK_{th} tends to the constant value $\Delta K_{th,lc}$. However, the fatigue strength prediction can be achieved using the same concepts underlying the $\sqrt{\text{area}}$ parameter model [31,38,39].
6. A reliable estimation of the fatigue strength needs a reliable estimation of the maximum defects occurring in a piece. This can be done by applying the statistics of extremes to the analysis of defects.
7. There is a number of open problems for the correct estimation of extreme defects: (i) a general method for the calculation of control volume in 2-D polished sections; (ii) the definition of upper limits for the use of type I distribution; (iii) the choice of the minimum control area (or volume) to be adopted.

Acknowledgments

This paper is a development both of the fruitful discussions we had during our visit to Department of Mathematics of Chalmers University of Technology and of our presentations at Symposium on “Statistical Methods on Fatigue of Materials”. We thank the Committee Chairman, Prof. Jacques De Maré, for giving us these opportunities.

Notes

1. The region where ΔK_{th} data are constant is essentially the region where linear elastic fracture mechanics (LEFM) is valid. This zone depends on the material and approximately corresponds to stress levels lower than $1/3\sigma_{wo}$ [7].
2. The slight difference from equation (5) comes from the fact that the SIF of a defect just in touch with the surface is higher than the SIF of a surface defect with the same size.
3. It is well known that fatigue strength tends to decrease with increasing dimensions of the specimens. This “size effect” is due to two principal factors: 1. increasing the dimensions of a specimen reduces the stress gradient in the more stressed areas and this effect is detrimental; 2. since fatigue failure always originates from defects or weak microstructures, the larger is the stressed volume the higher is the probability of failure. The latter factor, which was firstly evidenced by Weibull, depends on the material and it is referred to as statistical size effect [43,46].

References

- [1] Murakami, Y. and Endo, T., "Effect of small defects on the fatigue strength of metals," *Int. J. Fatigue* 2, 23–30, (1980).
- [2] Murakami, Y. and Endo, M., "Effect of hardness and crack geometries on ΔK_{th} of small cracks emanating from small defects," In: *The Behavior of Short Fatigue Cracks*, (K.J. Miller and E.R. de Los Rios, Eds.), MEP, London, 275–293, (1986).
- [3] Murakami, Y. and Endo, M., "Effect of defects, inclusions and inhomogeneities on fatigue strength," *Int. J. Fatigue* 16, 163–181, (1994).
- [4] Murakami, Y., *Metal Fatigue: Effects of Small Defects and Nonmetallic Inclusions*, Yokendo Ltd, Tokyo, 1993.
- [5] Kitagawa, H. and Takahashi, S., "Applicability of fracture mechanics to very small cracks or the cracks in the early stage," *Proc. 2nd Int. Conf. Mech. Behavior Mater—ICM2*, Boston, 627–631, (1976).
- [6] Murakami, Y., "Analysis of Stress Intensity Factors of modes I, II and III for inclined surface cracks of arbitrary shape," *Engng. Fract. Mech.* 22, 101–114, (1985).
- [7] Miller, K.J., "The short crack problem," *Fatigue Fract. Engng. Mater. Struct.* 5, 223–232, (1982).
- [8] Toriyama, T. and Murakami, Y., "The $\sqrt{\text{area}}$ parameter model for evaluation of effects of various artificial defects and mutual interaction of small defects at the fatigue limit," *J. Soc. Mater. Sci. Japan* 42, 1160–1166, (1993).
- [9] Murakami, Y., Toriyama, T., Koyasu, Y., and Nishida, S., "Effects of chemical composition of nonmetallic inclusions on the fatigue strength of high strength steels," *Tetsu-to-Hagane, Iron and Steel Inst. of Japan* 79, 678–684, (1993).
- [10] Murakami, Y., Kodama, S., and Konuma, S., "Quantitative evaluation of effects of non-metallic inclusions on fatigue strength of high strength steels I: basic fatigue mechanism and evaluation of correlation between the fatigue fracture stress and the size and location of non-metallic inclusions," *Int. J. Fatigue* 11(5), 291–298, (1989).
- [11] Duckworth, W.E. and Ineson, E., "The effects of externally introduced alumina particles on the fatigue life of En24 steel," *Clean Steel, Iron Steel Inst. Spec. Rep.* 77, 87–103, (1963).
- [12] Murakami, Y. and Usuki, H., "Quantitative evaluation of effects of non-metallic inclusions on fatigue strength of high strength steels II: fatigue limit evaluation based on statistics for extreme value of inclusion size," *Int. J. Fatigue* 11(5), 299–307, (1989).
- [13] Murakami, Y., Toriyama, T., and Coudert, E.M., "Instructions for a new method of inclusion rating and correlation with the fatigue limit," *J. Testing. Eval.* 22, 318–326, (1994).
- [14] Murakami, Y., "Inclusion rating by statistics of extreme value and its application to fatigue strength prediction and quality control of materials," *J. Res. Natl. Standard* 99, 345–351, (1994).
- [15] Gumbel E.J., *Statistics of Extremes*, Columbia University Press, New York, 1957.
- [16] Murakami, Y., *Basic Mechanisms in Fatigue of Metals*, Materials Science (P. Lukàs and J. Polàk Eds.), Elsevier, New York, 343–350, 1988.
- [17] Murakami, Y., "Effect of small defects and nonmetallic inclusions on the fatigue strength of metals," *JSME Int. J., Ser. A* 32, 167–180, (1989).
- [18] Murakami, Y., "Effects of nonmetallic inclusions on the fatigue strength of metals," In: *Micromechanics and Inhomogeneity*, The Toshio Mura 65th Anniversary Volume (G. J. Weng, M. Taya and H. Abe, Eds.), Springer-Verlag, New York, 283–295, (1990).
- [19] Murakami, Y., "Effects of nonmetallic inclusions of the fatigue of high strength steels: historical overview and new quantitative evaluation," *Impact of Improved Material Quality on Properties, Product Performance, and Design, MD.* 28, ASME, 89–103, (1991).
- [20] Kobayashi, M., Murakami, Y., Makino, T., Toriyama, T., Kurihara, Y., and Ebara, R., "Fatigue strength prediction of automobile suspension spring steels: explicit analysis of nonmetallic inclusions, shoot peening, decarburized layer, surface roughness and corrosion pits," *ibid.*, 171–183, (1991).
- [21] Natsume, Y., Miyakawa, S., Vemura, Y., and Murakami, Y., "Prediction of fatigue strength of high-speed steel based on estimation of the maximum size of inclusions," *Proc. Int. Conf. Fatigue 90*, (H. Kitagawa and T. Tanaka Eds.), Honolulu, 1, 349–354, (1990).

- [22] Murakami, Y., "Effects of small defects, small cracks and nonmetallic inclusions on fatigue strength of high strength steels," *Proc. Int. Conf. Fatigue 90*, In: (H. Kitagawa and T. Tanaka Eds.), Honolulu, 1, 377–382, (1990).
- [23] Endo, M., "Effects of graphite shape, size and distribution on the fatigue strength of spheroidal graphite cast irons," *J. Soc. Mater. Sci. Japan* 38(433), 1139–1144, (1989).
- [24] Sakai, T., Kogo, S., and Suzuki, M., "Statistical fatigue properties of nodular cast iron in rotating bending," *J. Soc. Mater. Sci. Japan* 38(434), 1268–1274, (1989).
- [25] Endo, M., "Fatigue thresholds for small cracks in spheroidal graphite cast iron," *Proc. Int. Conf. 'Fatigue 90'*, (H. Kitagawa and T. Tanaka Eds.), Honolulu, 3, 1357–1362, (1990).
- [26] Doi, S., Harada, S., Mitsunaga, K., Yano, M., and Yasuda, H., "Fatigue behavior of spheroidal graphite ductile iron," *Proc. JSME Meeting*, (900-86), 613–615, (1990).
- [27] Endo, M., "Fatigue strength prediction of nodular cast iron containing small defects," *Impact of Improved Material Quality on Properties, Product Performance, and Design*, MD 28, ASME, 125–137, (1991).
- [28] Sugiyama, Y., Asami, K., and Matsuoka, S., "The quantitative evaluation method on fatigue limit of materials with defects by using ductile cast iron and its application to the fatigue limit improvement," *Proc. 69th JSME Spring Ann. Meeting A(920-17)*, 476–478, (1992).
- [29] Endo, M., "Fatigue strength prediction of spheroidal graphite cast irons based on the root-area parameter model," *ibid.* 479–481, (1992).
- [30] Kato, Y. and Hasegawa, N., "Fatigue limits and thresholds in fatigue cracks growth of a spheroidal cast iron FCD450," *ibid.* 494–496, (1992).
- [31] Beretta, S., Blarasin, A., Endo, M., Giunti, T., and Murakami, Y., "Defect tolerant design of automotive components," *Int. J. Fatigue* 19, 319–333, (1997).
- [32] Kobayashi, M. and Matsui, T., "Prediction of fatigue strength of aluminum casting alloys by the $\sqrt{\text{area}}$ parameter model," *Trans. JSME, ser. A* 62(594), 341–346, (1996).
- [33] Murakami, Y., Ikeda, H., and Toriyama, T., "Quantitative evaluation of effects of inhomogeneity phases on the fatigue strength of Al-Si new alloys," *Proc. Int. Conf. Mech. Behavior Mater.—ICM6*, Kyoto (Japan), 433–438, (1991).
- [34] Natsume, Y., Muramatsu, T., and Miyamoto, T., "Effect of carbide crack on fatigue strength of alloy-tool steel under cold working," *Proc. JSME Meeting* (900-86) 323–325, (1990).
- [35] Beretta, S. and Murakami, Y., "Statistical analysis of defects for fatigue strength prediction and quality control of materials," *Fatigue Fract. Engng. Mater. Struct.* 21, 1049–1065, (1998).
- [36] Uemura, Y. and Murakami, Y., "A numerical simulation of evaluating the maximum size of inclusions to examine the validity of the metallographic determination of the maximum size of inclusions," *Trans. JSME* 56, 162–167, (1990).
- [37] Takahashi, R. and Shibuya, M., "The maximum size of the planar sections of random spheres and its application to metallurgy," *Ann. Inst. Statist. Math.* 48(1), 127–144, (1996).
- [38] Beretta, S. and Matteazzi, S., "Short cracks propagation in eutectoid steel wires," *Int. J. Fatigue* 18, 451–456, (1996).
- [39] Beretta, S. and Boniardi, M., "Fatigue strength and surface quality of high strength eutectoid steel wires," *Int. J. Fatigue* 21, 329–335, (1999).
- [40] Gladman, T., "Developments in inclusion control and their effects on steel properties," *Ironmaking and Steelmaking* 19(6), 457–463, (1992).
- [41] Wanibe, Y., Itoh, T., Umezawa, K., Nagahama, H., and Nori, Y., "Application of new techniques for characterization of non-metallic inclusions," *J. Steel Research* 66, 172–177, (1995).
- [42] Cybo, J., "Stereological alterations of non-metallic inclusions, fracture morphology and alloy impact strength after blowing active powders into liquid steel," *J. Steel Research* 66, 167–171, (1995).
- [43] Harter, L., "Size effect in fatigue," *Application of Statistics*, (P.R. Krishnaiah Ed.), North-Holland, 202–221, (1977).
- [44] Kohler, J. and Heckel, K., "Scatter and statistical size-effect at fatigue behavior of metallic specimens," *Proc. Int. Conf. Mech. Behavior Mater.—ICM2*, Boston (USA), 652–656, (1976).
- [45] Heckel, K. and Kra Ch., "Scatter and transferability of fatigue data at service loading," *Proc. Int. Conf. Mech. Behavior Mater.—ICM6*, Kyoto (Japan), 627–632, (1991).

- [46] Beretta, S., Clerici, P., and Paolini, G., “An experimental approach to size effect in rotating bending fatigue,” *Proc. Symp. Fatigue Design FD’92*, Helsinki (Finland), 122–133, (1992).
- [47] Bomas, H., Linkewitz, T., and Mayr, P., “Application of a weakest-link concept to the fatigue limit of hard steels,” *Proc. Symp., “Statistical methods in fatigue”*, Boras (Sweden). (1998).
- [48] Takahashi, R. and Shibuya, M., “Prediction of maximum size in Wicksell’s corpuscle problem,” *Ann. Inst. Statist. Math* 50(2), 361–377, (1998).
- [49] Anderson, C.W., “The largest inclusion within a piece of steel,” *Proc. Symp., “Extremes-risk and safety”*, Boras (Sweden) 1998.
- [50] Beretta, S. and Murakami, Y., “A new series of software for the analysis of defects based on the statistics of extremes,” *Report to NIRECO Ltd*, Kyushu University, Fukuoka (Japan) 1999.
- [51] Beretta, S. and Murakami, Y., “A new application of LEVD to the analysis of multiple inclusion types and the cleanliness assessment in metals,” To be submitted to *Acta Materialia*, (1999).